Supplementary Material for:

P-RP-PIL: A Novel Patch-Based Random Projection Pseudo-Inverse Learning Framework for Automated Retinal Disease Detection in Fundus Images

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This supplementary document contains extended analyses, secondary tables, and additional visualizations referenced in the main manuscript. All raw data and source code to reproduce these results are available at: https://github.com/your-username/p-rp-pil.

Supplementary Tables

Table 1: Sensitivity of P-RP-PIL to the random projection dimension L on both benchmarks (16×16 patches, raw patch dimension $D = 768$; *no projection* = raw patches fed directly to the hidden layer). Test-set accuracy (%) at seed 42. Production setting $L = 256$ is marked in bold.

Configuration	Papilledema (%)	DR (%)
No projection ($D = 768$)	92.23	83.12
Projection $L = 128$	94.17	88.31
Projection $L = 256$	**94.66**	**90.04**
Projection $L = 384$	94.17	85.71
Projection $L = 512$	88.35	87.01
Projection $L = 768$	93.20	83.12

Table 2: Multi-seed evaluation of P-RP-PIL over 10 independent runs (seeds 0–9). Each seed regenerates the random projection matrix and the hidden-layer weights and biases. For Papilledema, the train/test split is also regenerated per seed by pooling all images and re-splitting 70/15/15; for DR, the split is held fixed.

	Papilledema (10 seeds)			DR (10 seeds)		
Metric	Mean	SD	95% CI	Mean	SD	95% CI
Accuracy (%)	92.86	1.17	[92.03, 93.70]	91.30	1.39	[90.30, 92.29]
Precision	0.929	0.012	[0.920, 0.937]	0.913	0.014	[0.903, 0.923]
Recall	0.929	0.012	[0.920, 0.937]	0.913	0.014	[0.903, 0.923]
Specificity	0.959	0.005	[0.955, 0.963]	0.913	0.014	[0.903, 0.923]
F1-score	0.928	0.012	[0.920, 0.936]	0.913	0.014	[0.903, 0.923]
AUC	0.983	0.007	[0.978, 0.988]	0.937	0.019	[0.924, 0.950]

Table 3: Sensitivity to test-time perturbations. Values are percentage-point drops relative to clean accuracy; negative values indicate a slight improvement. Gaussian noise ($\sigma = 0.1$) and +20% brightness shifts were applied to the test set only. Lowest drop per column in bold.

	Papilledema		DR	
Model	Noise	Brightness	Noise	Brightness
P-RP-PIL	64.56	3.88	37.23	0.43
DenseNet-121	13.11	0.00	46.32	0.43
ResNet-50	17.48	6.31	17.75	−1.30
MobileNetV2	34.47	3.40	5.19	−0.87
EfficientNet-B0	19.90	0.49	22.94	−0.43
Custom-CNN	**11.65**	−0.97	**0.43**	−0.43

Table 4: Empirical sweep of patch size and stride on the Papilledema dataset. Accuracy plateaus at patch size 8; stride dominates training time without affecting accuracy. The production configuration (patch 8, stride 2) is marked in bold.

Patch size	Stride	Accuracy (%)	Time (s)
4	1	88.83	22.5
4	2	88.83	23.8
8	**1**	**94.17**	22.0
8	**2**	**94.17**	**21.1**
8	4	94.17	7.1
12	1	92.72	25.5
12	2	92.72	23.9
12	4	90.78	8.1
16	1	93.20	26.6
16	2	93.20	27.1
16	4	94.66	8.3

Table 5: Empirical sweep of patch size and stride on the DR dataset. Accuracy is highest at patch size 4 with stride 1 (93.07%) but at 93.2 s of reduced-pipeline time, whereas the production configuration (patch 6, stride 3) achieves 92.64% in 9.5 s; a 9.8× speed-up for a 0.43 pp accuracy cost. The reduced-pipeline timings are not directly comparable to the full-pipeline time of 12.6 s in Table III of the main text.

Patch size	Stride	Accuracy (%)	Time (s)
4	1	**93.07**	93.2
4	2	90.91	18.8
4	3	92.21	9.4
6	1	92.64	92.3
6	2	90.91	19.5
6	**3**	**92.64**	**9.5**
6	4	90.04	6.8
8	1	90.04	95.1
8	2	90.04	18.9
8	3	91.77	11.3
8	4	91.34	7.6
12	1	90.04	104.4
12	2	90.91	21.9
12	3	90.04	11.3
12	4	89.61	7.3
16	1	90.91	102.2
16	2	91.77	27.1
16	3	90.04	13.2
16	4	90.04	7.6

Table 6: Component-wise ablation on the DR dataset. Each row flips exactly one component relative to the production configuration (see §IV.A of the main text). Test N = 231. The bottom two rows replace the closed-form pseudo-inverse with a linear and an RBF SVM trained on the same hidden features.

Configuration	Accuracy (%)	Δ (pp)
Full P-RP-PIL	**92.64**	—
w/o attention weighting	92.21	−0.43
w/o random projection	90.91	−1.73
w/o PCA	89.18	−3.46
with multi-scale patches	87.88	−4.76
with Gabor features	87.01	−5.63
with binary hashing	81.39	−11.26
with Random Fourier Features	83.98	−8.66
with graph-Laplacian	92.64	0.00
w/o PIL (linear SVM on hidden features)	93.94	+1.30
w/o PIL (RBF SVM on hidden features)	93.07	+0.43

Table 7: Extended Metrics with 95% Confidence Intervals (DR Dataset), from the 10-seed evaluation of Table S2.

Metric	P-RP-PIL
Accuracy (%)	91.30 ± 1.39
Sensitivity	0.913 ± 0.014
Specificity	0.913 ± 0.014
F1-score	0.913 ± 0.014
AUC	0.937 ± 0.019

Supplementary Figures

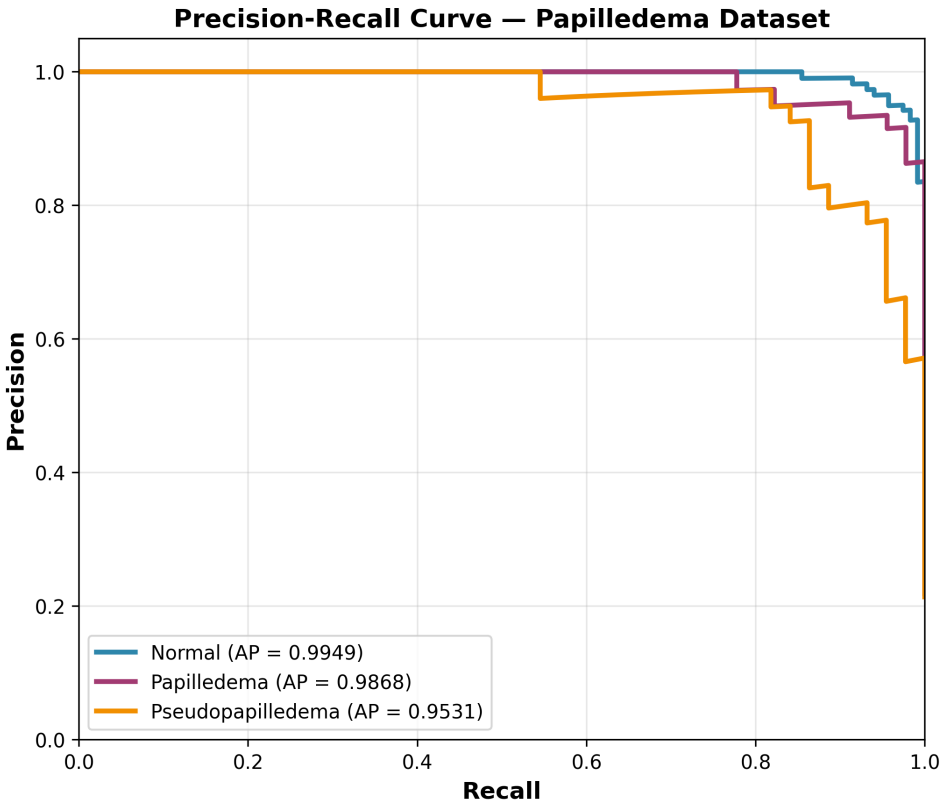


Figure 1: Precision-Recall curves for the Papilledema dataset. Papilledema AP = 0.987, Normal AP = 0.995, Pseudopapilledema AP = 0.953.

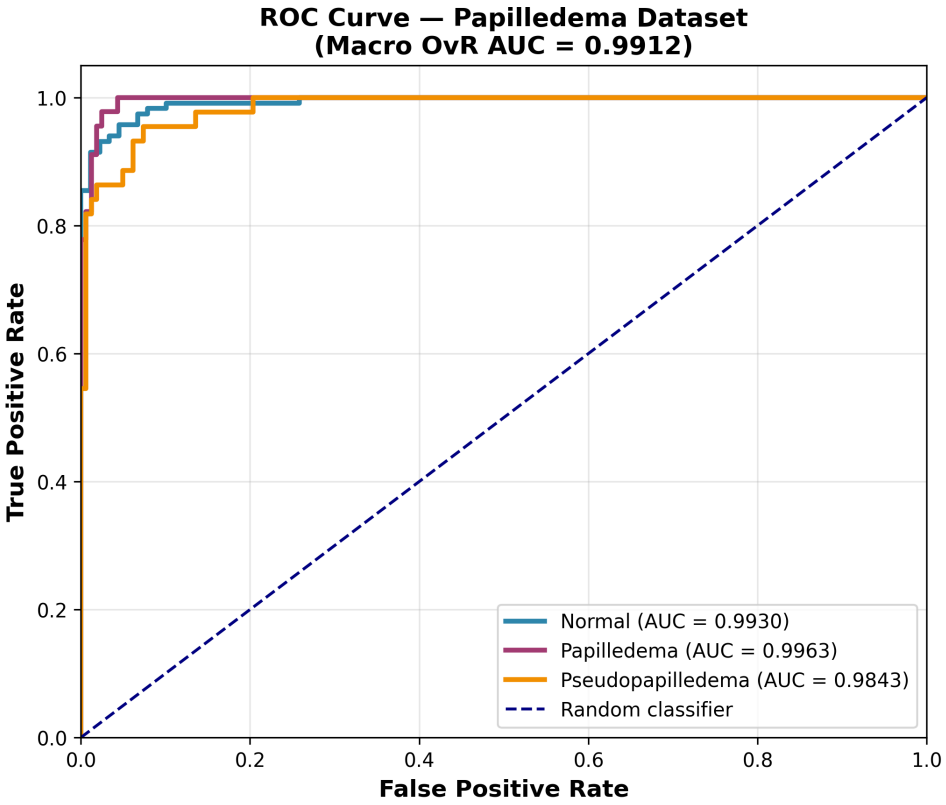


Figure 2: ROC curves for the Papilledema dataset (one-vs-rest). Papilledema AUC = 0.9963, Normal AUC = 0.9931, Pseudopapilledema AUC = 0.9843. Macro one-vs-rest AUC = 0.9912.

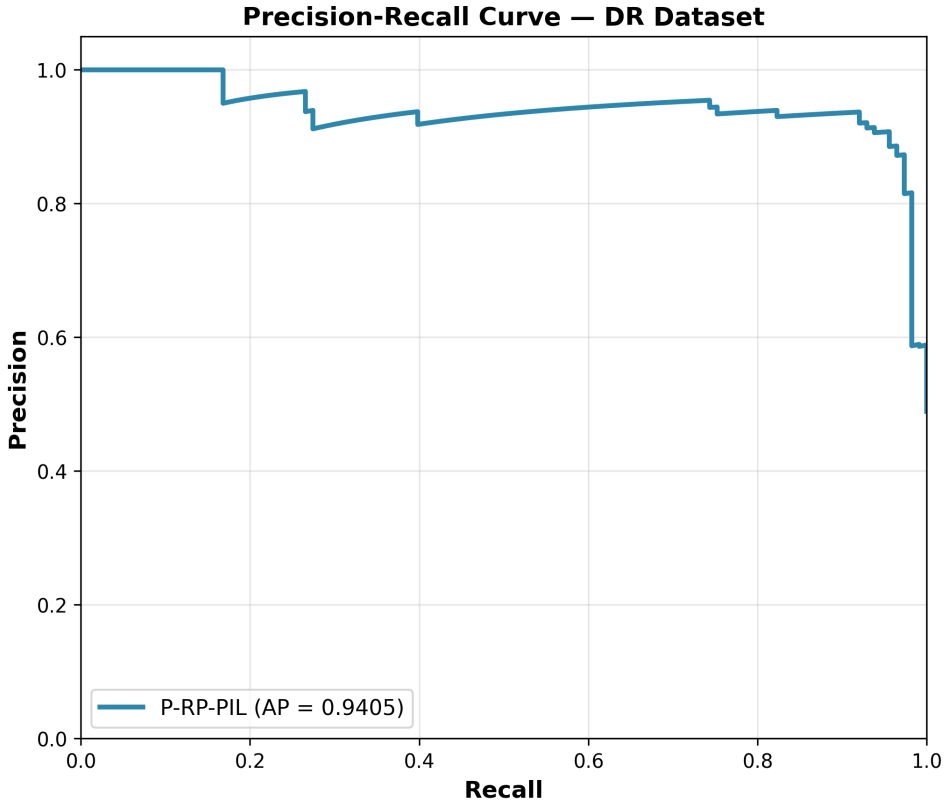


Figure 3: Precision-Recall curve for the DR dataset. P-RP-PIL achieves AP = 0.9405.

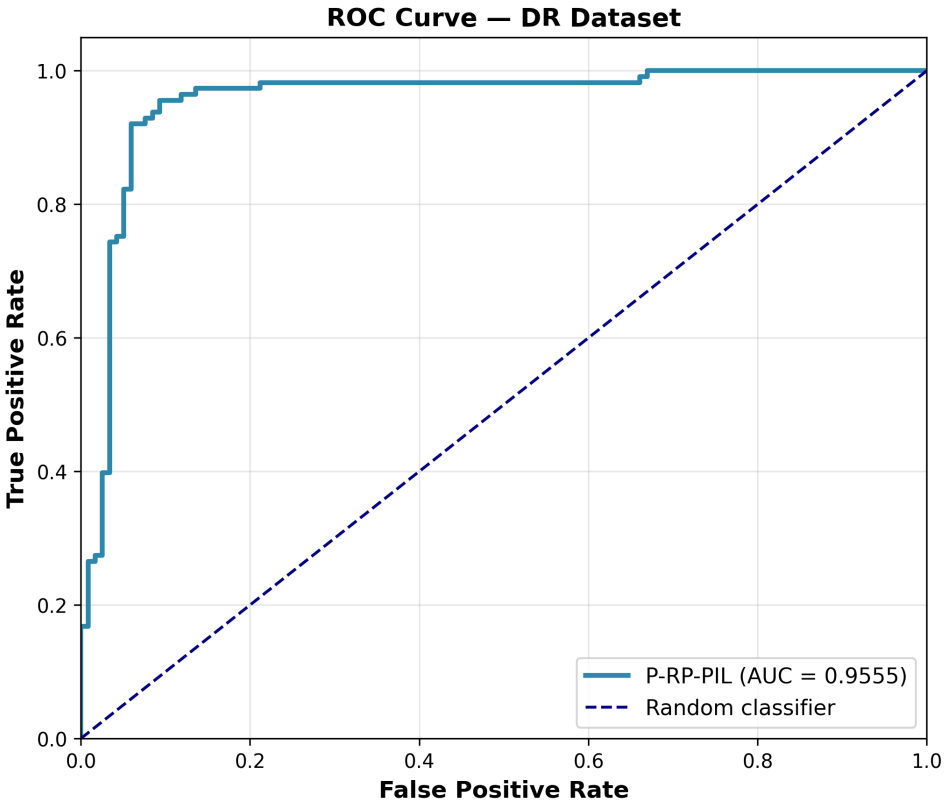


Figure 4: ROC curve for the DR dataset (single-run evaluation). P-RP-PIL achieves AUC = 0.9555. The 10-seed mean AUC is 0.937 ± 0.019 (Table S7).

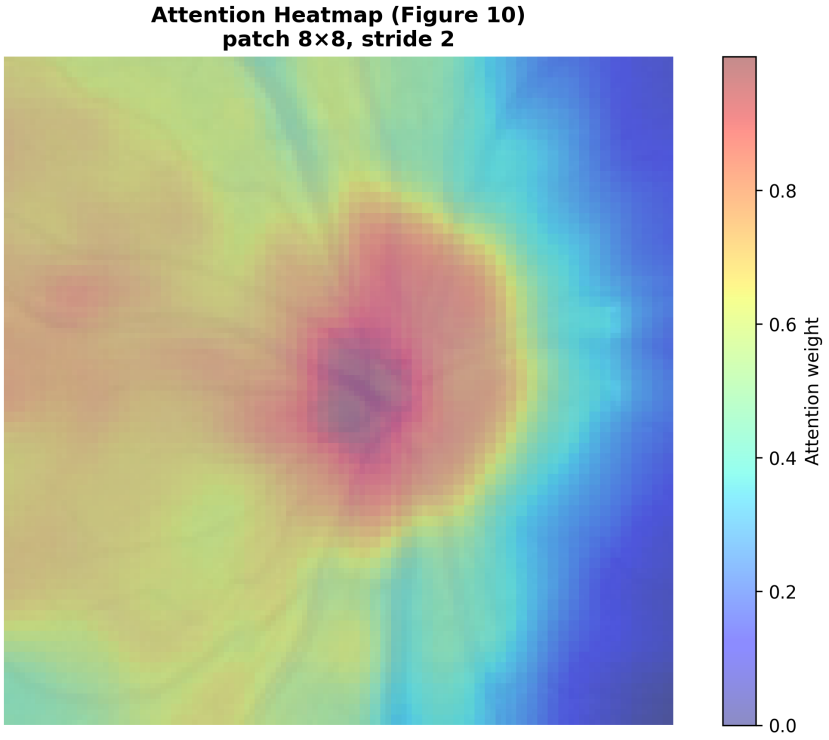


Figure 5: Attention heatmap for a representative papilledema case, computed from the model's own variance-based per-patch weights.

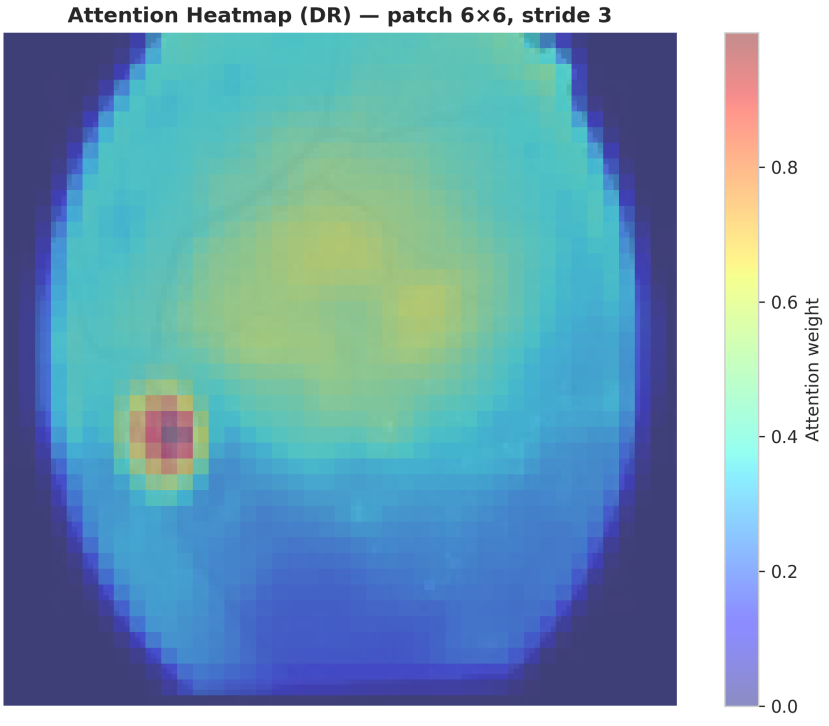


Figure 6: Attention heatmap for a representative DR case, computed from the model's variance-based per-patch weights.

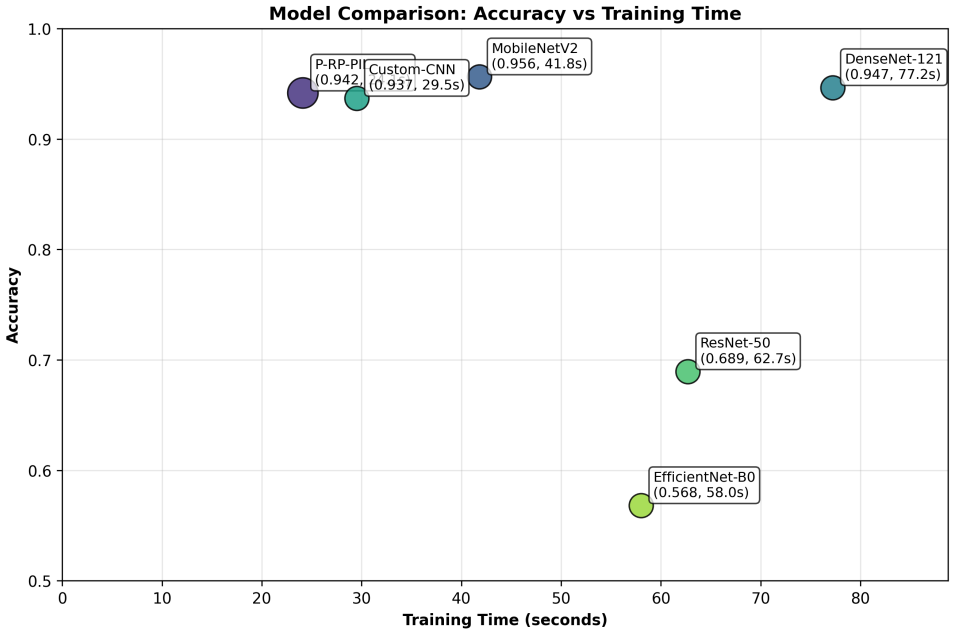


Figure 7: Accuracy versus training time for all evaluated models on the Papilledema dataset. P-RP-PIL lies on the efficiency frontier.

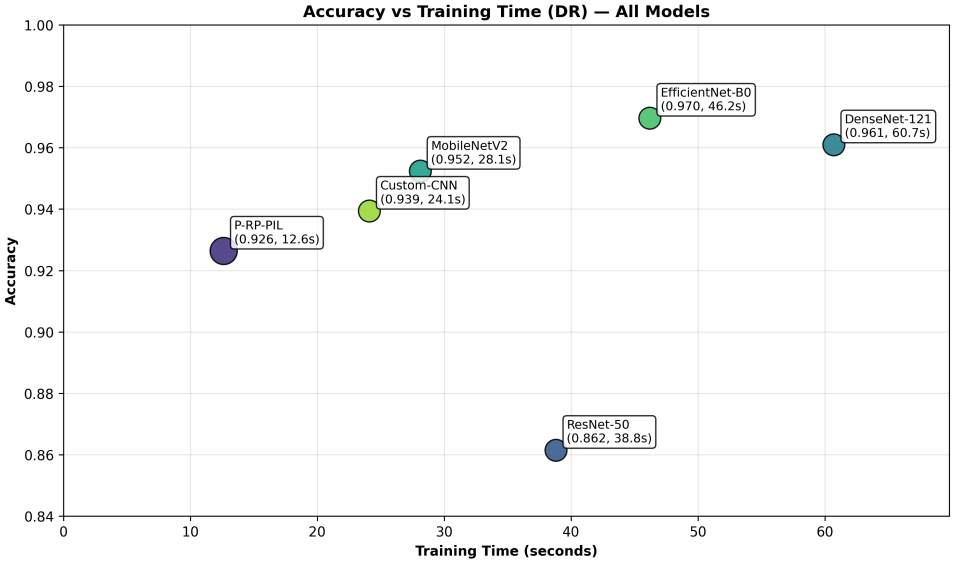


Figure 8: Accuracy versus training time for all six models on the DR dataset. P-RP-PIL is the fastest model in the comparison, training in 12.6 s — 4.8× faster than the highest-accuracy model (EfficientNet-B0, 46.2 s) and 4.8× faster than DenseNet-121 (60.7 s).

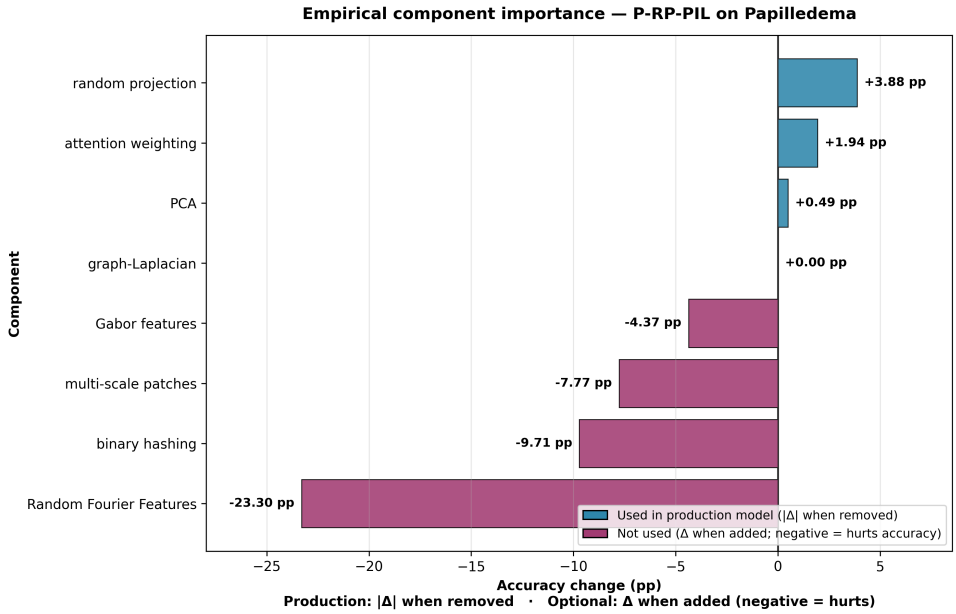


Figure 9: Empirical component importance for P-RP-PIL on Papilledema. Blue bars: accuracy drop (Δ in pp) when a production component is removed from the full model. Red bars: accuracy change (Δ in pp) when an optional component is added; negative values indicate the component hurts accuracy. All values are computed from the ablation study of Table V.

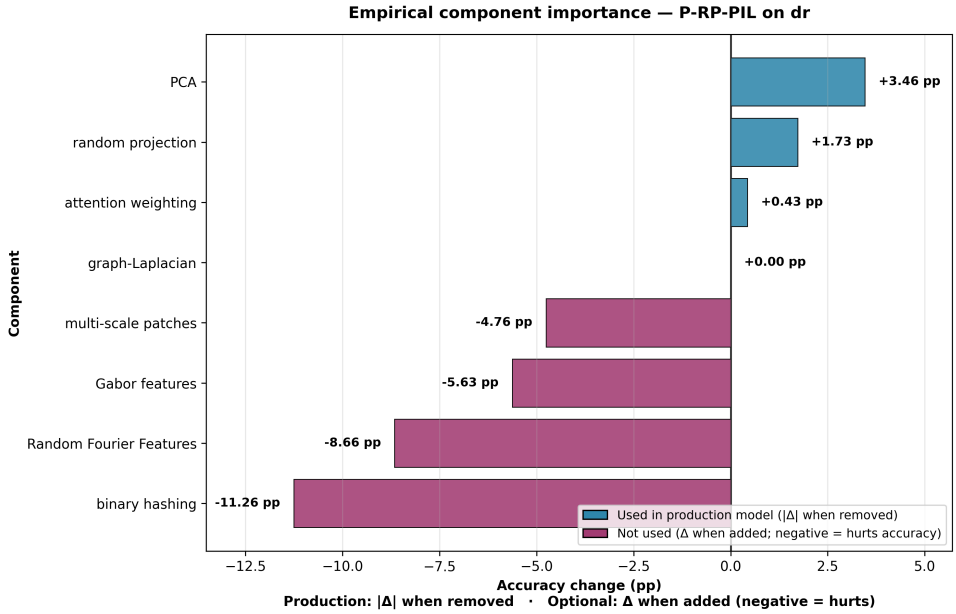


Figure 10: Empirical component importance for P-RP-PIL on DR. Blue bars: accuracy drop (Δ in pp) when a production component is removed. Red bars: accuracy change (Δ in pp) when an optional component is added; negative values indicate the component hurts accuracy. All values are computed from the ablation study of Table S6.